

Abdullah Haneef

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EDUCATION

Delhi Technological University

Delhi, India

B.Tech. in Computer Engineering – CGPA: 9.03/10.00

Dec 2021 – Jul 2025

- **Bachelor Thesis:** *Decoding Pneumonia: Evaluating Convolutional Neural Networks and Transformer Architectures Using Integrated Gradients* (Advisor: Prof. Rahul Katarya)
- **Competitive Exams:** JEE Advanced Rank 9716 (Top 1.5% of ~1.5M candidates); JEE Mains Rank 6256 (Top 0.6%).
- **English Proficiency:** TOEFL iBT Score = 109/120

RELEVANT COURSEWORK

Mathematics & Theory: Calculus I-II, Discrete Mathematics, Probability & Statistics, Logical Reasoning

Computer Science: Artificial Intelligence, Machine Learning, Computer Vision, Data Structures & Algorithms, Computer Organization & Architecture, Computer Networks, Compiler Design, Operating Systems, Distributed Systems, DBMS, OOPS

PUBLICATIONS

A. Haneef, A. Ahmad, R. Katarya. “Decoding Pneumonia: Evaluating Convolutional Neural Networks and Transformer Architectures Using Integrated Gradients.” *Proceedings of the ICDAM (Springer)*, 2025. [[Link to Paper](#)]

RESEARCH EXPERIENCE

Mitralab, Ashoka University

Delhi, India

Research Assistant (Advisor: Prof. Partha Pratim Chakrabarti, Prof. Kasturi Mitra)

Oct 2025 – Present

- Developing a U-Net based pipeline to quantify mitochondrial morphology in human induced pluripotent stem cells (WTC-11).
- Implementing watershed segmentation to automate organelle analysis, targeting a 30-50% reduction in manual annotation time.

AIISC, University of South Carolina

Remote, USA

AI Researcher (Advisor: Prof. Amitava Das)

Sep 2024 – Sep 2025

- Developed defense mechanisms for text-to-image models against adversarial perturbations using Patch-based Optimization.
- Achieved a 65% recovery in model accuracy against PGD and FGSM attacks by regularizing the latent space representation.

Unmanned Aerial Systems Laboratory, DTU

Delhi, India

AI Research Intern (Advisor: Prof. S. Indu)

Aug 2024 – Aug 2025

- Improved the semantic segmentation of off-road terrains by integrating Diffusion optimization into the U-Net bottleneck.
- Refined object boundary detection in unstructured environments, increasing Mean IoU (mIoU) by 4% compared to the baseline.

Machine Learning Laboratory, DTU

Delhi, India

AI Research Intern (Advisor: Prof. Anil Parihar)

Jul 2023 – Feb 2024

- Simulated Vision-Language-Navigation (VLN) agents in the Matterport3D environment.
- Incorporated Multi-View Geometry constraints to resolve object occlusion, increasing navigation Success Rate (SR) by 12%.

DTU Self-Driving Car Project

Delhi, India

ML Research & Perception Lead (Advisor: Prof. Qasim Murtaza)

Dec 2021 – Jul 2023

- Architected a sensor fusion pipeline integrating Velodyne LiDAR and ZED 2 Stereo Camera for 3D object detection, achieving 88% mAP@0.5.
- Implemented Extended Kalman Filters (EKF) for SLAM, reducing localization error by ~10cm RMSE in GPS-denied environments.
- Deployed the perception stack on NVIDIA Jetson AGX using ROS2/C++.

WORK EXPERIENCE

BlockVerse Institute

Remote, Netherlands

Artificial Intelligence Intern

Feb 2024 – Jul 2024

- Engineered "Insight Hire," a multimodal interviewing system fusing visual cues with audio features extracted via Wav2Vec.
- Implemented real-time proctoring algorithms to detect anomalies, achieving a false positive rate of < 5% during live sessions.

ACADEMIC SERVICE

Reviewer: International Journal of Electrical and Computer Engineering (IJECE) (Scopus Indexed).

SELECT PROJECTS

Image-Caption Generator | *PyTorch, Inception v3, LSTM*

- Implemented an Encoder-Decoder architecture using Inception v3 for feature extraction and LSTM for caption generation.
- Achieved a BLEU-4 score of 26.5 on the Flickr8k test split. [\[Code\]](#)

Video Interview Analysis Tool | *OpenCV, OpenPose, FER*

- Developed a CV pipeline detecting micro-expressions and posture deviations using FER and OpenPose algorithms.
- Deployed via Streamlit; optimized frame processing to run at 30 FPS on CPU. [\[Code\]](#)

TECHNICAL SKILLS

Programming: Python, C++, SQL, Bash, $\text{\LaTeX}$

Frameworks: PyTorch, TensorFlow, ROS/ROS2, OpenCV, HuggingFace Transformers, Docker

Research Interests: Computer Vision, Multimodal Learning, Robot Perception, Generative AI

Concepts: SLAM, Sensor Fusion, Graph Neural Networks, Markov Decision Processes

AWARDS & ACHIEVEMENTS

Formula Student Italy (2022): Won 2nd prize in Business Plan; Ranked 8th overall among 200 international teams.

Execute 2.0 Hackathon (2022): Top 10 Finalist among 100+ teams; built a proximity-based competitive platform for street vendors.